Exc : 11
Date : 10/9/26

Routing at network layer

a) Simulates statics Routing protocol configuration using CISCO Packet Tracer.

Aim :-
To configure static routing between different network using cisco packet Tracer and verify communication between the network.

steps :-

1) Open cisco packet Tracker
2) place 2 routers, 2 switches and PC's
3) connect PC's to switches and switches to routers.
4) Assign IP address to all PC's and router interfaces.
5) configure static routers using the command
6) configure the router interface and enable the using shoutdown.
7) configure static routers on both router.
8) set the appropriate default gate way on each PC.
9) use the Ping command to test communication between different network.
10) use Simulation mode to observe the packet path.

Result :-

Hence the static routing was successfully configured between the network using cisco packet tracer and communication between the network was successfully verified using the ping command.

output:-

Router0

Router show ip route static

S 30.0.0.018 [01 0] via 20.0.0.2

S 30.0.0.10013 2 [1010] via 40.0.0.2

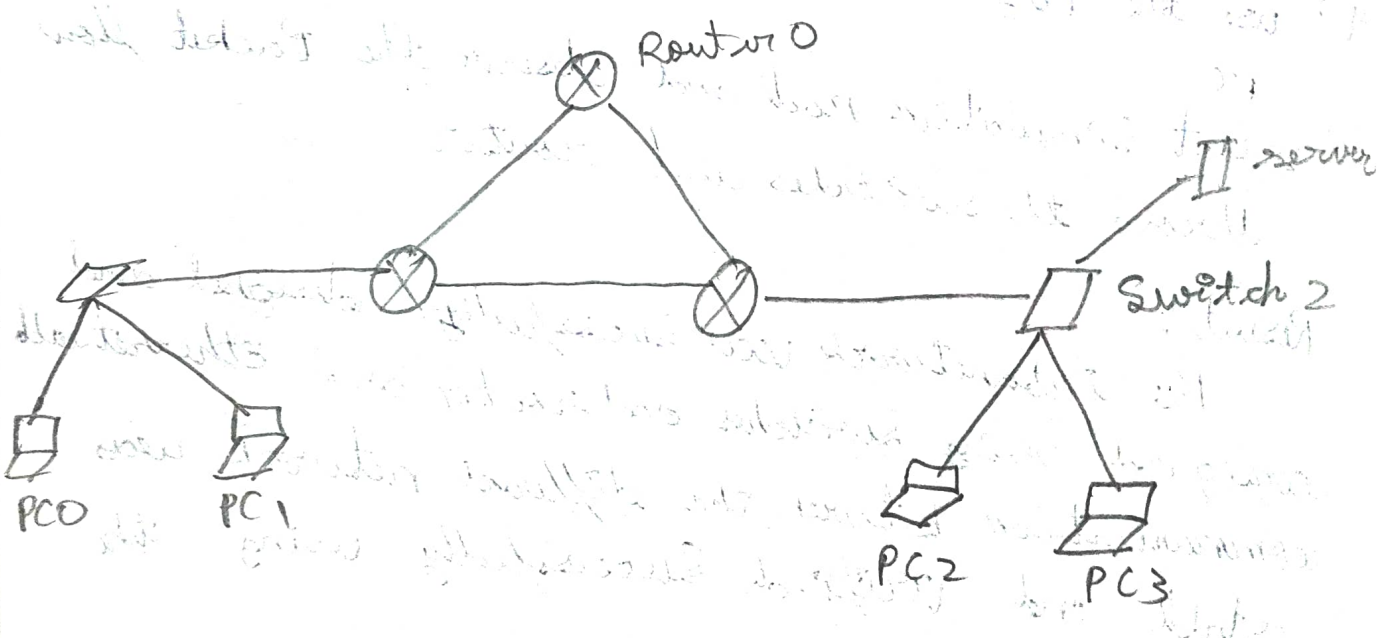
S 50.0.0.018 [10.10] via 40.0.0.2

Router 1

Router # show ip route static

S 10.0.0.018 [10.10] via 20.0.0.1

S 40.0.0.0.18 [10.10] via 10.0.0.1



11) b) Simulate RIP using cisco packet tracer.
10/9/26

Aim :-

To configure and simulate the Routing Information protocol (RIP) between routers using cisco packet Tracer and verify network connectivity.

steps :-

1) Open cisco packet Tracer
2) place 2 or 3 routers, Switches and PCs
3) connect the devices using ethernet cables.
4) Assign IP addresses to PC's and router interfaces.
5) Enable all router interfaces using no shutdown
6) Configure RIP on the Routers.
7) Configure the appropriate network addresses on each router according to the topology.
8) check the routing table using :- show ip route.
9) use the ping command to test communication between PC's in different networks.
10) use Simulation mode to observe how RIP learns and elect routers.

Result :-

RIP was successfully configured between the routers using cisco packet Tracer. The routes learned the required network router dynamically and communication between different networks was successfully verified.

Ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data

Request times out

Reply from 10.0.0.2 bytes 32 time=3ms TTL=126

Reply from 10.0.0.2 bytes=32 time=3ms TTL=126

Reply from 10.0.0.2 bytes=32 time=3ms TTL=126

Ping statistics for 10.0.0.2

packets = sent=4, Received=3, lost=1

Approximate round trip in milli-seconds

minimum = 3ms, Maximum = 3ms,

Average = 3ms